

ARYABHATTA KNOWLEDGE UNIVERSITY (AKU), PATNA

BCA — 3rd Semester

BCA-302

INTERNET AND WEB DESIGNING

Code: 303302

Semester	BCA 3rd Semester
Full Marks	60 Marks
Duration	3 Hours
Pattern	7 questions. Attempt 5. Q1 & Q2 compulsory.
Marks Pattern	Q1: 6×2=12 Q2: 3×4=12 Q3-Q7: 3×12=36
PYQ Years	2014, 2017, 2018, 2019, 2020, 2022, 2024, 2025 (ALL VERIFIED from uploaded papers)
Syllabus Source	Official AKU syllabus image uploaded by student (BCA-302 / 303302)

QUESTIONS COVERED IN THIS BOOKLET

- UNIT 1 — Internet Basics — Basic concepts, TCP/IP, Web Server/Client, Domain Registration, IP Address, Protocols
- UNIT 2 — Introduction to HTML — HTML Tags, Text Formatting, Lists, Tables, Images, Links, Frames
- UNIT 3 — JavaScript — Data Types, Operators, Arrays, Functions, Conditional Checking, Type Casting
- UNIT 4 — Understanding XML — SGML, XML, XML vs HTML, XML Structure, XML Schema
- UNIT 5 — Dynamic Web Pages using JSP — Dynamic Web Page, JSP Scripting, Standard Actions, Page Directive, Include Directive

SOURCE NOTE: All PYQ years are verified ONLY from uploaded question paper PDF (2014–2025). Syllabus from uploaded image. No year or question has been invented.

UNIT 1 — INTERNET BASICS

TCP/IP | Web Server | Domain | IP
Address | Protocols

Q1

What is TCP/IP? Explain in detail the layers of TCP/IP protocol and its services.

12 Marks

PYQ: 2017, 2018,
2019, 2020*PYQ Evidence: Repeated in 2017 (Q3), 2018 (Q3), 2019 (Q6), 2020 (Q3) — most repeated 12-mark question.*

TCP/IP (Transmission Control Protocol / Internet Protocol) is the communication protocol suite that forms the backbone of the Internet. It defines how data is packaged, sent, and received between devices on a network.

TCP/IP is a set of communication protocols used to interconnect network devices on the Internet. TCP ensures reliable, ordered data delivery; IP handles addressing and routing of packets across networks.

TCP/IP 4-Layer Model — A T I N:

1. **Application Layer (Top):** Provides network services to end-user applications. Protocols: HTTP (web browsing), SMTP (email), FTP (file transfer), DNS (domain names), Telnet (remote login), SNMP.
2. **Transport Layer:** Provides end-to-end data delivery. TCP (reliable, connection-oriented, ordered, error-checked) or UDP (unreliable, connectionless, fast — for video/gaming).
3. **Internet Layer:** Handles logical addressing (IP addresses) and routing of packets. IPv4 = 32-bit addressing. IPv6 = 128-bit addressing. Finds best path across networks.
4. **Network Access Layer (Bottom):** Physical data transmission over network medium. Protocols: Ethernet, Wi-Fi (802.11), Token Ring. Handles MAC addresses.

TCP/IP 4-LAYER MODEL:

```

+-----+
| APPLICATION LAYER (HTTP, FTP, SMTP, DNS) |
+-----+
| TRANSPORT LAYER      (TCP, UDP)          |
+-----+
| INTERNET LAYER       (IP, ICMP, ARP)      |
+-----+
| NETWORK ACCESS LAYER (Ethernet, Wi-Fi)   |
+-----+

```

TCP/IP Services:

1. File Transfer (FTP): Transfer files between computers using port 21.
2. Email (SMTP/POP3/IMAP): Send (SMTP port 25) and receive (POP3 port 110) electronic mail.
3. Remote Login (Telnet/SSH): Log into and control remote computers.
4. Web Browsing (HTTP/HTTPS): Access web pages on port 80 (HTTP) and 443 (HTTPS).
5. Domain Name Service (DNS): Converts domain names (www.google.com) to IP addresses.
6. Network Management (SNMP): Monitor and manage network devices.

TCP/IP is the universal language of the Internet. Its 4-layer architecture provides reliability (TCP), global addressing (IP) and flexibility for all types of network communication.

MEMORY TRICK: TCP/IP Layers (top to bottom): ATIN = Application, Transport, Internet, Network Access. TCP = reliable. UDP = fast. IPv4 = 32-bit. IPv6 = 128-bit.

Q2**What is the basic purpose of domain registration? Explain the working of electronic mail.****12 Marks**

PYQ: 2022

Domain registration gives websites a unique, memorable address. Electronic mail (email) enables text, file and multimedia exchange over the Internet.

Domain Registration: The process of reserving a unique domain name (e.g., akupatna.ac.in) through an authorized registrar for a specific period. It links a human-readable name to an IP address via DNS.

1. Purpose of Domain Registration: Provides unique identity to a website, makes it easy to find without memorizing IP addresses, establishes brand identity online.
2. Domain Extensions (TLDs): .com (commercial), .org (organization), .edu (education), .gov (government), .in (India), .net (network)
3. Registration Steps: Choose domain name → Check availability with registrar (GoDaddy, Namecheap) → Pay fee → Domain linked to web server IP via DNS → Domain is live.

Working of Electronic Mail (Email):

1. Compose: User writes email in Mail Client (Gmail, Outlook) with To, CC, Subject and Body.
2. Send (SMTP): Mail client sends email to sender's Mail Server using SMTP (port 25).
3. DNS Lookup: Sender's Mail Server looks up recipient domain's Mail Exchanger (MX) record via DNS.
4. Transfer: Email transferred to recipient's Mail Server using SMTP.
5. Storage: Email stored on recipient's Mail Server until retrieved.
6. Receive (POP3/IMAP): Recipient's mail client downloads email using POP3 (port 110) or IMAP (port 143).

EMAIL FLOW:

```
[Sender's PC] → SMTP → [Sender's Mail Server]
      ↓ DNS Lookup
[Recipient's Mail Server] ← SMTP ←
      ↓ POP3/IMAP
[Recipient's PC]
```

Domain registration gives websites a unique address on the Internet. Email works through SMTP for sending and POP3/IMAP for receiving, passing through multiple mail servers.

MEMORY TRICK: Domain: Choose → Check → Pay → DNS links → Live. Email: SMTP to send. POP3/IMAP to receive. DNS to find the right server.

UNIT 2 — INTRODUCTION TO HTML

Tags | Formatting | Lists | Tables | Forms | Frames | Links

Q3

What is HTML? Explain the structure of an HTML document. Describe HTML tags, commonly used commands, text formatting and lists.

12 Marks
PYQ: 2019, 2020, 2024, 2025

PYQ Evidence: Repeated 2019 (Q4), 2020 (Q4), 2024 (Q4), 2025 (Q4) — consistently most asked 12-mark HTML question.

HTML is the standard language for creating web pages. Every website you visit is built with HTML at its foundation. It defines the structure and content using a system of tags.

HTML (HyperText Markup Language) is a standard markup language used to create and structure web pages. It uses tags enclosed in angle brackets to define elements. HTML is not a programming language — it is a markup language that structures content.

Structure of HTML Document:

```
<!DOCTYPE html>      ← Declares HTML5 document
<html>                ← Root element
  <head>               ← Metadata (not displayed)
    <title>My Page</title> ← Browser tab title
    <meta charset='UTF-8'> ← Character encoding
  </head>
  <body>               ← Visible content
    <h1>Main Heading</h1>
    <p>This is a paragraph.</p>
    <a href='page2.html'>Click Here</a>
    <img src='photo.jpg' alt='Photo'>
  </body>
</html>
```

Important HTML Tags:

Tag	Purpose	Example
<h1>-<h6>	Headings (h1=largest)	Heading
<p>	Paragraph	Text here
	Hyperlink	Click
	Insert image (self-closing)	
 	Line break (self-closing)	End of line New line
<hr>	Horizontal rule	<hr>
/	Bold text	Bold
<i>/	Italic text	<i>Italic</i>
<u>	Underline	<u>Underline</u>
<!-- -->	Comment (not displayed)	<!-- This is a comment -->

HTML Lists — 3 Types:

```
<!-- Unordered list (bullets) -->
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<!-- Ordered list (numbers) -->
<ol>
  <li>First</li>
  <li>Second</li>
</ol>

<!-- Definition list -->
```

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
</dl>
```

HTML Table Structure:

```
<table border='1' cellpadding='5'>
  <tr>
    <th>Name</th>    <!-- th = bold header cell -->
    <th>Age</th>
  </tr>
  <tr>
    <td>Abdul</td>    <!-- td = data cell -->
    <td>21</td>
  </tr>
</table>
```

HTML provides the skeleton of every web page. Structure: DOCTYPE → html → head + body. Head has metadata. Body has visible content. HTML works with CSS for styling and JavaScript for interactivity.

MEMORY TRICK: HTML structure: D-H-H-B = DOCTYPE, HTML (root), HEAD (metadata), BODY (content). Lists: ul=bullets, ol=numbers, dl=definitions. Table: table → tr → th/td.

Q4**What is HTML form? Explain the elements of HTML form with example.****12 Marks**

PYQ: 2018

HTML forms are the primary way to collect user input on web pages. Login forms, registration forms, search boxes — all are HTML forms.

An HTML form is an element used to collect user input. Data entered is sent to a server using the form's action attribute. The method attribute specifies GET (visible in URL) or POST (hidden, secure).

1. <input type='text'>: Single-line text field. Attribute name= sends data. Placeholder= shows hint text.
2. <input type='password'>: Password field — input masked with * or dots.
3. <input type='radio'>: Radio buttons. Same name= groups them; only one can be selected.
4. <input type='checkbox'>: Checkboxes. Multiple options can be selected simultaneously.
5. <input type='submit'>: Button to submit form data to the server.
6. <input type='reset'>: Button to clear all form fields back to defaults.
7. <textarea>: Multi-line text area. Attributes: rows= and cols= set size.
8. <select> + <option>: Dropdown list. name= attribute used to retrieve selected value.

```
<form action='process.jsp' method='POST'>
  Name: <input type='text' name='name'><br>
  Password: <input type='password' name='pwd'><br>
  Gender:
    <input type='radio' name='g' value='M'> Male
    <input type='radio' name='g' value='F'> Female<br>
  Hobby:
    <input type='checkbox' name='h' value='Music'> Music
    <input type='checkbox' name='h' value='Sports'> Sports<br>
  Course:
    <select name='course'>
      <option>BCA</option>
      <option>MCA</option>
    </select><br>
  Address:<br>
  <textarea name='addr' rows='3' cols='30'></textarea><br>
  <input type='submit' value='Register'>
  <input type='reset' value='Clear'>
</form>
```

HTML forms are the main way to collect user data for server processing. GET method shows data in URL (for search). POST method hides data (for passwords, forms).

MEMORY TRICK: Form method: GET=visible in URL (search), POST=hidden (login/register). Always use POST for passwords! action= is where data goes. name= is the key to retrieve data on server.

UNIT 3 — JAVASCRIPT

Data Types | Operators | Arrays |
Functions | Events

Q5

What is JavaScript? Explain various operators and data types available in JavaScript with examples.

12 Marks

PYQ: 2020, 2022

PYQ Evidence: 2020 Q6 (operators), 2020 Q5 (JS advantages), 2022 Q7 (operators and data types).

JavaScript makes web pages interactive and dynamic. It runs directly in the browser without any installation. Every modern website uses JavaScript for user interaction.

JavaScript is a lightweight, client-side scripting language used to create dynamic and interactive web pages. Created by Brendan Eich in 1995. It runs in the browser, is interpreted (not compiled), and is case-sensitive.

Data Types in JavaScript — 7 Types:

Data Type	Description	Example
Number	Integers and decimals	var n=25; var pi=3.14;
String	Text in single or double quotes	var s='Abdul'; var c="Patna";
Boolean	True or false values only	var f=true; var done=false;
Undefined	Declared variable without value	var x; // x is undefined
Null	Intentionally empty/no value	var y=null;
Object	Collection of key-value pairs	var obj={name:'Ali',age:21};
Array	Ordered list of values	var arr=[10,20,30];

Operators in JavaScript — 6 Types:

Type	Operators	Example	Result
Arithmetic	+ - * / % ++ --	5 + 3 10 % 3	8 1
Comparison	== === != !== > < >= <=	5==5 5===5 5==='5'	true true true(loose)
Logical	&& (AND) (OR) ! (NOT)	true&&false; true false	false true
Assignment	= += -= *= /= %=	x=5; x+=3	x=8
String	+ (concatenation)	'Hello'+'World'	'HelloWorld'
Ternary	condition ? val1 : val2	(5>3) ? 'yes' : 'no'	'yes'

```
<script>
  var a=10, b=3;
  document.write(a+b);    // 13 arithmetic
  document.write(a>b);    // true comparison
  document.write(a&&b);    // true logical
  var x = (a>b) ? a : b;   // 10 ternary
  var n1=5, n2=3;
  document.write(++n1);    // 6 pre-increment
  document.write(n2++);    // 3 post-increment (then n2=4)
</script>
```

Advantages and Disadvantages of JavaScript:

Advantages	Disadvantages
Client-side execution — reduces server load	Security risk — code visible to all users
No compilation — runs directly in browser	Browser compatibility issues may occur
Interactive UI — form validation, animations	JavaScript can be disabled by user
Supports all browsers — cross-platform	Debugging can be difficult
Rich ecosystem — jQuery, React, Node.js	Single-threaded — no true multitasking

JavaScript is essential for modern web development. It provides client-side interactivity, form validation and dynamic content without server round-trips. Its operators and data types are similar to Java/C but more flexible.

MEMORY TRICK: 7 Data types: Number, String, Boolean, Undefined, Null, Object, Array. JS is LOOSELY TYPED — variable type can change. 5=='5' is TRUE (loose), 5==='5' is FALSE (strict).

Q6

What is a JavaScript function? Explain user-defined functions with examples. Explain how to declare and access elements of an array in JavaScript.

12 Marks

PYQ: 2017, 2019

Functions and arrays are two of the most important concepts in JavaScript. Functions organise code into reusable blocks; arrays store and manage collections of data.

A JavaScript function is a named block of code that performs a specific task. It is defined once and can be called many times with different arguments, producing different results.

1. Function without parameters: void-type function that just executes code on call.
2. Function with parameters: accepts input values for processing.
3. Function with return value: returns a result that can be stored in a variable.
4. Anonymous function: function without a name, stored in a variable.

```
// 1. Without parameters
function greet() {
    document.write('Hello World!');
}
greet(); // call

// 2. With parameters
function add(a, b) {
    document.write(a + b);
}
add(5, 3); // Output: 8

// 3. With return value
function multiply(a, b) {
    return a * b;
}
var result = multiply(4, 5);
document.write(result); // Output: 20

// 4. Function called via event
// <button onclick='greet()'>Click Me</button>
```

JavaScript Arrays:

1. Declaration: var arr = [10, 20, 30]; or var arr = new Array(10, 20, 30);
2. Index starts from 0: arr[0]=10, arr[1]=20, arr[2]=30
3. Array methods: push() add to end | pop() remove from end | length get size | sort() sort | reverse() reverse | concat() join arrays | join() join as string

```
var fruits = ['Apple', 'Mango', 'Banana'];
document.write(fruits[0]);    // Apple
document.write(fruits.length); // 3

fruits.push('Grape');          // ['Apple', 'Mango', 'Banana', 'Grape']
fruits.pop();                  // removes 'Grape'
fruits.sort();                 // ['Apple', 'Banana', 'Mango']
var a=[1,2]; var b=[3,4];
var c = a.concat(b);           // [1,2,3,4]
document.write(fruits.join(', ')); // Apple, Banana, Mango
```

Functions make code reusable and organized. Arrays efficiently store and manage multiple values. Together, they form the foundation of JavaScript programming.

MEMORY TRICK: Function types: No params, With params, With return, Anonymous. Array: index from 0. Methods: push(add), pop(remove), sort(), concat(join arrays), length(size).

UNIT 4 — UNDERSTANDING XML

SGML | XML | XML vs HTML |
Structure | Schema

Q7

What is the difference between SGML and XML? Write XML syntax for storing Book details of 5 books in a bookstore.

12 Marks

PYQ: 2019

PYQ Evidence: 2019 Q4 (12M) — SGML vs XML with bookstore XML. Also related: 2018 Q4a (XML structure).

SGML is the parent language from which both HTML and XML were derived. XML simplified SGML for web use while maintaining its power for data representation.

SGML (Standard Generalized Markup Language) is an ISO standard meta-language for defining markup languages, developed in 1986. XML (eXtensible Markup Language) is a simplified subset of SGML designed for web and data exchange, standardized by W3C in 1998.

Aspect	SGML	XML
Year	1986 (ISO standard)	1998 (W3C standard)
Complexity	Very complex	Simple and easy
DTD	Mandatory	Optional (XSD preferred)
Closing tags	Optional for some elements	All tags MUST be closed
Case sensitivity	Not case-sensitive	Case-sensitive (<Name> ≠ <name>)
Use case	Large publishing systems	Web, APIs, data exchange
Derived from	Parent of HTML and XML	Subset of SGML
Self-describing	No	Yes

XML Rules (Must follow ALL):

1. Must have exactly one root element that contains all other elements.
2. All elements must have a closing tag — <book>...</book> or self-closing
3. XML is case-sensitive — <Book> and <book> are different elements.
4. All attribute values must be quoted — id="1" not id=1
5. Tag names cannot have spaces — use <bookTitle> not <book title>

XML for Bookstore — 5 Books:

```
<?xml version='1.0' encoding='UTF-8'?>
<bookstore>
  <book id='1'>
    <title>HTML and XML for Beginners</title>
    <author>Michel Morrison</author>
    <publisher>PHI</publisher>
    <price>450</price>
    <year>2002</year>
  </book>
  <book id='2'>
    <title>Java How to Program</title>
    <author>H.M. Deitel</author>
    <publisher>PHI</publisher>
    <price>650</price>
    <year>2005</year>
  </book>
  <book id='3'>
    <title>JavaScript Definitive Guide</title>
    <author>David Flanagan</author>
    <publisher>O Reilly</publisher>
    <price>750</price>
    <year>2020</year>
  </book>
  <book id='4'>
```

```
<title>Web Design Complete Reference</title>
<author>Thomas Powell</author>
<publisher>McGraw Hill</publisher>
<price>550</price>
<year>2001</year>
</book>
<book id='5'>
  <title>JSP Programming</title>
  <author>WROX Team</author>
  <publisher>WROX Publication</publisher>
  <price>500</price>
  <year>2003</year>
</book>
</bookstore>
```

XML is a platform-independent way to store and exchange structured data. Unlike HTML (which displays data), XML describes and stores data. Its strict rules ensure data integrity across systems.

MEMORY TRICK: XML rules: 1 Root, All closed, Case-sensitive, Quoted attributes. XML=data storage. HTML=data display. SGML=parent of both. Well-formed XML = follows syntax rules.

UNIT 5 — DYNAMIC WEB PAGES USING JSP**JSP Scripting | Standard Actions |
Page Directive | Include Directive****Q8****What is JSP? Explain its advantages. Explain JSP standard actions in detail with syntax.****12 Marks**PYQ: 2014, 2017,
2018, 2020

PYQ Evidence: 2014 Q6 (JSP actions), 2017 Q6 (JSP advantages), 2018 Q6 (JSP actions with syntax), 2020 Q7 (JSP advantages).

JSP (JavaServer Pages) is a server-side technology that enables developers to create dynamic, database-driven web pages by embedding Java code directly into HTML. It is part of the Java EE platform.

JSP (JavaServer Pages) is a technology that enables the creation of dynamic web content by embedding Java code in HTML pages using special tags. JSP pages are translated to Servlets by the JSP engine and run on a web server.

How JSP Works:

```

Client Browser → Request .jsp page
                ↓
        Web Server/JSP Engine
                ↓
        1. Translate JSP → Servlet (.java)
        2. Compile Servlet → Bytecode (.class)
        3. Execute Servlet → Generate HTML
                ↓
        HTML Response → Client Browser (displays page)
  
```

Advantages of JSP:

1. Dynamic Content: Generates different HTML based on user input, database results or session data.
2. Java Power: Full access to Java libraries — OOP, exception handling, JDBC for database.
3. Platform Independent: Runs on any OS with JVM + web server (Tomcat, JBoss).
4. Separation of Concerns: Presentation (HTML) and logic (Java) in same file but distinct sections.
5. Reusable Components: JavaBeans and custom tags promote code reuse.
6. Built-in Objects: request, response, session, out etc. available automatically.
7. Error Handling: errorPage attribute for centralized error management.

JSP Scripting Elements — SED:

Element	Syntax	Purpose	Example
Scriptlet	<% ... %>	Java code (statements, logic)	<% int x=5; out.println(x); %>
Expression	<%= ... %>	Outputs value to browser	<%= new java.util.Date() %>
Declaration	<%! ... %>	Declare variables and methods	<%! int count=0; %>
Directive	<%@ ... %>	Instructions to JSP engine	<%@ page import='java.util.*' %>
Comment	<%-- ... --%>	JSP comment (not sent to client)	<%-- JSP comment --%>

JSP Standard Actions — 6 Main Actions:

1. <jsp:include page='header.jsp'/> — Dynamically includes content of another page at request time. The included page is processed each time.
2. <jsp:forward page='next.jsp'/> — Forwards the current request to another resource (JSP, Servlet or HTML). Stops current page processing.
3. <jsp:useBean id='b' class='UserBean' scope='session'/> — Creates or finds a JavaBean. scope= can be: page, request, session or application.
4. <jsp:setProperty name='b' property='*'/> — Sets property/properties of a JavaBean. property='*' auto-maps form fields to bean properties.
5. <jsp:getProperty name='b' property='name'/> — Retrieves and displays a property value from a JavaBean.

6. `<jsp:param name='key' value='val'/>` — Passes parameters to included or forwarded pages.

JSP Page Directive — All Important Attributes:

```
<%@ page language='java'
    contentType='text/html; charset=UTF-8'
    import='java.util.*, java.io.*'
    session='true'
    errorPage='error.jsp'
    buffer='8kb'
    extends='parentClass'
%>
```

Note: 'Export' is NOT a valid page directive attribute (PYQ 2019)

JSP Lifecycle — 6 Phases:

1. Translation → JSP file → Servlet .java source
2. Compilation → .java → bytecode .class file
3. Loading → Class loaded into JVM memory
4. `jspInit()` → Called ONCE on first request (initialization)
5. `_jspService()` → Called for EACH request (main processing)
6. `jspDestroy()` → Called ONCE when JSP is removed (cleanup)

JSP Implicit Objects (9 — Automatically Available):

Object	Type	Purpose
request	HttpServletRequest	Get client data: <code>getParameter()</code> , <code>getHeader()</code>
response	HttpServletResponse	Send data back to client
out	JspWriter	Write output to browser (<code>out.println()</code>)
session	HttpSession	Store user data across requests
application	ServletContext	Data shared across all users/sessions
config	ServletConfig	Servlet configuration parameters
pageContext	PageContext	Access all other implicit objects
page	Object	Reference to current JSP page (like 'this')
exception	Throwable	Available only in error pages

JSP is a powerful technology for building dynamic web applications. Its scripting elements (SED: Scriptlet, Expression, Declaration), standard actions and implicit objects provide complete control over dynamic page generation.

MEMORY TRICK: JSP scripting: SED = Scriptlet(<% %>), Expression(<%= %>), Declaration(<%! %>). 6 lifecycle: Translate → Compile → Load → Init → Service → Destroy. Export is NOT a valid page directive attribute!

Q9

Write HTML and JSP codes for sending registration details (first name, last name, username, password) from HTML page and displaying them on a JSP page.

12 Marks

PYQ: 2018

This question combines HTML form design with JSP server-side processing — testing both front-end and back-end web development skills.

```
<!-- FILE 1: register.html -->
<html>
<head><title>Registration Form</title></head>
<body bgcolor='#f0f8ff'>
<center>
<h2>Student Registration Form</h2>
<form action='display.jsp' method='POST'>
  <table border='1' cellpadding='8' bgcolor='white'>
    <tr><td>First Name:</td>
      <td><input type='text' name='fname' required></td></tr>
    <tr><td>Last Name:</td>
      <td><input type='text' name='lname' required></td></tr>
    <tr><td>Username:</td>
```

```

        <td><input type='text' name='uname' required></td></tr>
    <tr><td>Password:</td>
        <td><input type='password' name='pwd' required></td></tr>
    <tr><td colspan='2' align='center'>
        <input type='submit' value='Register'>
        <input type='reset' value='Clear'></td></tr>
</table>
</form>
</center></body></html>

```

```

<!-- FILE 2: display.jsp -->
<%@ page language='java' contentType='text/html; charset=UTF-8' %>
<html>
<head><title>Registration Details</title></head>
<body bgcolor='#e8f5e9'>
<center>
<%
    String fname = request.getParameter('fname');
    String lname = request.getParameter('lname');
    String uname = request.getParameter('uname');
    // Note: password not displayed for security
%>
<h2>Registration Successful!</h2>
<table border='1' cellpadding='10' bgcolor='white'>
    <tr bgcolor='#4CAF50'><th>Field</th><th>Value</th></tr>
    <tr><td>First Name</td><td><%= fname %></td></tr>
    <tr><td>Last Name</td><td><%= lname %></td></tr>
    <tr><td>Username</td><td><%= uname %></td></tr>
    <tr><td>Password</td><td>***** (hidden)</td></tr>
    <tr><td>Full Name</td><td><%= fname+' '+lname %></td></tr>
</table>
<br><a href='register.html'>Back to Form</a>
</center></body></html>

```

HTML form sends data to JSP using action='display.jsp' method='POST'. JSP retrieves it using request.getParameter('fieldname'). Expressions <%= %> output values directly.

MEMORY TRICK: Complete flow: HTML form → POST → JSP → request.getParameter() → <%= output %>. Always use method='POST' for passwords. Never display password in plain text!

QUICK REVISION TABLE — All Questions + Key Points

PYQ Verified

Q#	Topic	Type	PYQ Years	One-Line Key Point
1	TCP/IP Layers & Services	12M	2017,2018,2019,2020	4 layers: ATIN. TCP=reliable, UDP=fast. IPv4=32-bit, IPv6=128-bit.
2	Domain Registration + Email	12M	2022	DNS converts domain → IP. Email: SMTP=send, POP3/IMAP=receive.
3	HTML Structure + Tags + Lists	12M	2019,2020,2024,2025	DOCTYPE → html → head+body. ul=bullets, ol=numbers, table → tr → td.
4	HTML Form Elements	12M	2018	input types: text,password,radio,checkbox,submit. form action+method.
5	JavaScript Data Types + Operators	12M	2020,2022	7 types: Number,String,Boolean,Undefined,Null,Object,Array. 6 ops: ACLAST.
6	JS Functions + Arrays	12M	2017,2019	Function: define once, call many. Array: index from 0, push/pop/sort/concat.
7	SGML vs XML + Bookstore XML	12M	2019	SGML=parent,complex. XML=simple,1 root,all tags closed,case-sensitive.
8	JSP Actions + Advantages	12M	2014,2017,2018,2020	Scripting: SED. Actions: include,forward,useBean,setProperty,getProperty.
9	HTML+JSP Registration Form	12M	2018	HTML form → POST → JSP → request.getParameter() → display.
MC Q1	Father of WWW	2M	2014	Tim Berners-Lee (not Dennis Ritchie!)
MC Q2	Primary web protocol	2M	2014	HTTP (not TCP/IP, not FTP)
MC Q3	IPv6 size	2M	2017,2019	128 bits (NOT 32, NOT 64)
MC Q4	Image tag	2M	2014,2017,2024,2025	 (not <image>)
MC Q5	Top JS object	2M	2014	window (not document)
MC Q6	Joins arrays	2M	2014	concat() (not join, not add)
MC Q7	Server-side technology	2M	2018	JSP (not HTML, CSS, JavaScript)
MC Q8	JSP stands for	2M	2025	JavaServer Pages (not JavaScript Pages)
MC Q9	JSP expression element	2M	2025	<%=...%> (not <% %> or <%! %>)
MC Q10	Export in page directive	2M	2019	Export is NOT valid (Language,Import,Session are valid)

FINAL CHEAT SHEET — Read Before Entering Exam Hall

UNIT 1: Internet Basics

Term	Full Form / Meaning
HTTP	HyperText Transfer Protocol — web browsing (port 80)
HTTPS	HTTP Secure (port 443)

Term	Full Form / Meaning
FTP	File Transfer Protocol (port 21)
SMTP	Simple Mail Transfer Protocol — send email (port 25)
POP3	Post Office Protocol — receive email (port 110)
DNS	Domain Name System — converts domain name to IP
ISP	Internet Service Provider — provides internet connection
TCP	Transmission Control Protocol — reliable, ordered delivery
UDP	User Datagram Protocol — fast, unreliable
IPv4	32-bit IP address IPv6 = 128-bit IP address

UNIT 2: HTML Quick Tags

Tag	Use	Remember
	Unordered list (bullets)	UL = Unordered
	Ordered list (numbers)	OL = Ordered
	List item	Used inside ul/ol
<tr>	Table row	tr = table row
<td>	Table data cell	td = table data
<th>	Table header (bold)	th = table header
	Insert image	img NOT image!
	Hyperlink	a = anchor
<frameset>	Replaces <body> for frames	cols= or rows=
<hr>	Horizontal rule line	hr = horizontal rule

UNIT 3: JavaScript Critical Facts

Fact	Answer
Created by / Year	Brendan Eich / 1995 at Netscape
Runs where?	Client-side (in browser). NOT on server.
How to output	document.write() or alert() or console.log()
Top-level object	window (contains document, history, location)
Array index starts from	0 (zero)
joins arrays	concat() — arr1.concat(arr2)
Sort array	arr.sort() or arr.sort(function(a,b){return a-b;})
moves window to position	moveTo(x,y)
Removes focus	blur() — removes focus from element/window
== vs ===	== loose (5=='5' is true). === strict (5==='5' is false)

UNIT 4: XML Rules

Rule	Details
One root element	XML must have exactly 1 root element containing all others
All tags must close	<book>...</book> or self-closing
Case-sensitive	<Book> and <book> are different elements!
Quote all attributes	id='1' or id="1" — never id=1
XML vs HTML	XML=store data, HTML=display data. XML=custom tags, HTML=fixed tags

Rule	Details
SGML	Parent of XML and HTML. Complex. 1986 ISO standard.
SAX	Event/callback parser — fast, memory efficient
DOM	Tree-based parser — loads entire XML into memory

UNIT 5: JSP Must-Know

Concept	Quick Answer
JSP stands for	JavaServer Pages — NOT JavaScript Pages!
Scriptlet	<% Java code %> — runs but doesn't output directly
Expression	<%= value %> — outputs value to browser page
Declaration	<%! variable or method %> — class-level definition
Directive	<%@ page/include/taglib %> — instructions to JSP engine
Export attribute	NOT valid in page directive! (this is a PYQ trap)
get form data	request.getParameter('fieldname')
jspInit()	Called ONCE when JSP is first loaded
_jspService()	Called for EACH HTTP request — main logic here
jspDestroy()	Called ONCE when JSP is removed from service
include directive	Static inclusion at translation time: <%@ include file=" %>
jsp:include	Dynamic inclusion at request time — processed each time

AKU EXAM WRITING FORMAT (For 12-Mark Questions): Introduction (2-3 lines) → Definition (in box) → Numbered Points (bold headings) → Diagram/Code → Example → Conclusion → Memory Trick. Time: Q1 MCQ=15 min | Q2 Short Notes=30 min | Q3-Q7 Long=25 min each. Total=3 hours.